

Professional Certificate Programme in Agentic AI and Applications



Agent Architectures & Collaboration



Revisit the Learning So Far

Let's take a quick look at the key ideas and concepts you have explored in your learning journey.

- Understanding the AI ecosystem, including agents, functions and APIs
- How the agent life cycle progresses from initial goals to continuous growth
- An overview of agentic frameworks such as LangChain, Autogen and CrewAI
- Core principles that guide tool integration within AI agent frameworks
- How AI agents can be extended and enhanced using external APIs

Topics Covered

- Agentic AI System Architectures
- Types of AI Agents: Fundamentals & Industry Applications
- Task-Oriented v. Goal-Oriented Agents & Agent Communication Frameworks

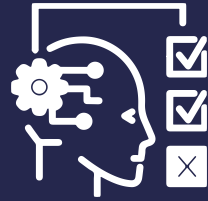
Agentic AI System Architectures

Agentic AI System Architectures



- Range from reactive responses to multi-agent orchestration
- Support specific use cases and complexity levels in modern AI systems

The Architecture Spectrum



Agentic AI
systems span a
spectrum of
complexity and
autonomy



Architectural
choices depend
on task
complexity,
autonomy level
and multi-agent
coordination
needs

The Architecture Spectrum



Simple & fast

Reactive patterns
for immediate
responses



Workflow focus

Deliberative
approaches for
complex decisions



Distributed intelligence

Multi-agent systems
for coordinated
tasks agents

Reactive Agents: The Thermostat Approach

Stimulus-Response Pattern



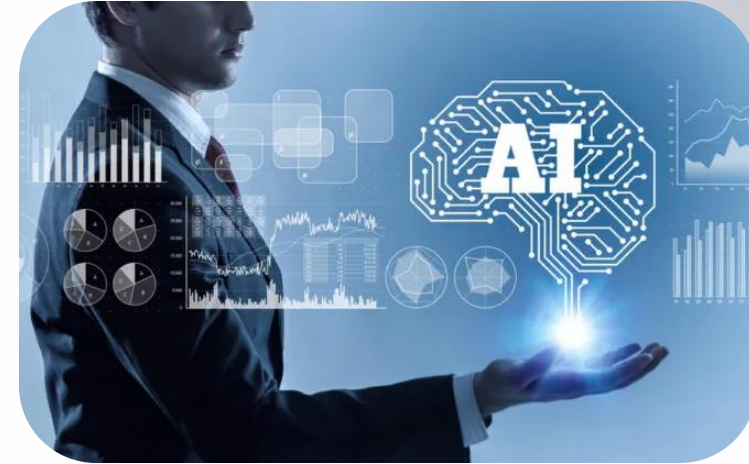
Best for: Real-time systems requiring instant responses with predictable patterns

- Operate on a simple principle: "If X happens → do Y."
- Do not store history or plan ahead, enabling fast, efficient reactions
- Ideal for real-time tasks, such as chatbots, trading bots and robotic sensors
- Excel at predictable responses, like thermostats adjusting to temperature changes

Deliberative Agents: The Chess Master



- Maintain internal models and plan actions using symbolic reasoning
- Excel in strategic scenarios, such as autonomous navigation and complex problem-solving



- Excel in tasks requiring foresight and strategic thinking
- Used in autonomous vehicles for route optimisation in expert systems for complex domains like medical diagnosis and financial planning

Trade-off: Deliberative agents sacrifice speed for sophistication, making slower but more informed decisions by considering context and long-term impact.

Hybrid Agents: Best of Both Worlds

Reactive layer

Instant responses to critical situations like:

- Emergency braking
- Collision avoidance
- Real-time adjustments

Deliberative layer

Strategic planning and reasoning like:

- Route optimisation
- Resource allocation
- Long-term goals

Hybrid Agents: Best of Both Worlds



- Self-driving cars combine deliberative planning for optimal routing with reactive responses to dynamic hazards
- This dual-layer design enables strategic foresight alongside rapid, instinctive action

BDI Agents: Human-Like Reasoning



Beliefs

Represents the agent's knowledge of the world and environment



Desires

Defines the objectives and goals the agent aims to achieve



Intentions

Tracks plans and actions committed to achieving those goals.

Mirror human reasoning, by updating beliefs, evaluating desires and committing to intentions

Act contextually, such as allocating resources when deadlines approach to ensure project success

Multi-Agent Systems: Team Intelligence



- Represent distributed intelligence, where multiple agents interact via established communication protocols
- Each agent maintains specialised capabilities, while contributing to collective objectives

Multi-Agent Systems: Team Intelligence



Cooperative

Agents collaborate toward shared goals, like drone swarms in search and rescue



Competitive

Agents compete for resources within rules, as seen in algorithmic trading



Coordinated

Agents perform specialised roles with orchestrated actions, like in supply chains

Learning & Tool-Using Agents

Learning agents

Improve performance through feedback loops, using reinforcement learning to refine decisions over time



- Continuously refine behaviour, like a piano student responding to feedback
- Applied in Game AI, adaptive tutoring and optimisation bots

Tool-Using agents

Extend capabilities by calling external APIs and functions, selecting the right tool for each task



- Leverage external resources intelligently, like a handyman choosing tools with precision
- Used in LangChain agents, API-driven assistants and computational tools

Architecture Decision Matrix

Architecture pattern	Speed	Complexity	Ideal use cases
Reactive	 Fast	 Simple	Real-time chatbots, trading bots, sensor systems
Deliberative	 Slow	   Complex	Strategic planning, knowledge systems, route optimisation
Hybrid	  Mixed	    Very complex	Autonomous vehicles, adaptive assistants, robotics
Multi-agent	 Fast	     Extremely complex	Supply chains, drone swarms, distributed systems

Key Architectural Insights

Real-world autonomous systems blend reactive speed with deliberative planning for optimal performance in dynamic settings

Modern agents extend capabilities via APIs and external tools, making tool-using patterns essential

Multi-agent orchestration enables horizontal scaling by distributing intelligence across specialised components

Hybrid is king



Tool integration



Scalability matters



Choose architecture by task needs, not trends—the best agent solves your problem effectively.

Types of AI Agents: Fundamentals & Industry Applications

Types of AI Agents



- Four core types of AI agents underpin real-world applications across diverse industries
- Architectures range from simple reactive systems to complex multi-agent orchestrations

Agent Architecture Overview

- Perceive their environment
- Take goal-directed actions

- System performance
- Scalability
- Complexity



Agent Architecture Overview



Environmental

perception: Agents collect data from sensors, APIs, databases or users to assess context.



Action execution: Agents act via API calls, robotic movements, data updates or system communication.



Decision processing:

Agents apply reactive rules, planning or hybrid logic to process information.



Learning & adaptation:

Advanced agents learn from outcomes to refine decisions and adapt to change.

Reactive Agents: Stimulus-Response Systems

Core Characteristics

Reactive agents operate on direct stimulus-response patterns without internal models or memory



- They execute predetermined actions based on "**if X then Y**" behavioural rules in response to environmental inputs:
 - No internal world model or memory
 - Immediate response to environmental stimuli
 - Simple rule-based decision making
 - Minimal computational overhead
 - Fast execution but limited adaptability

Key advantage: Speed and simplicity make reactive agents ideal for time-critical tasks requiring millisecond responses.

Reactive Agents in Industry

Error handling

- Assembly-line robots stop instantly when sensors detect obstacles
- No planning—just immediate response to sensor data

High-frequency trading

- Trading bots execute buy/sell orders in microseconds based on price movements
- Speed is everything—deliberation risks missed opportunities

Smart home systems

- Thermostats adjust temperature based on sensor readings
- Simple but effective — Heating activates immediately when temperature drops below threshold

Reactive Agents in Industry

Training Insight

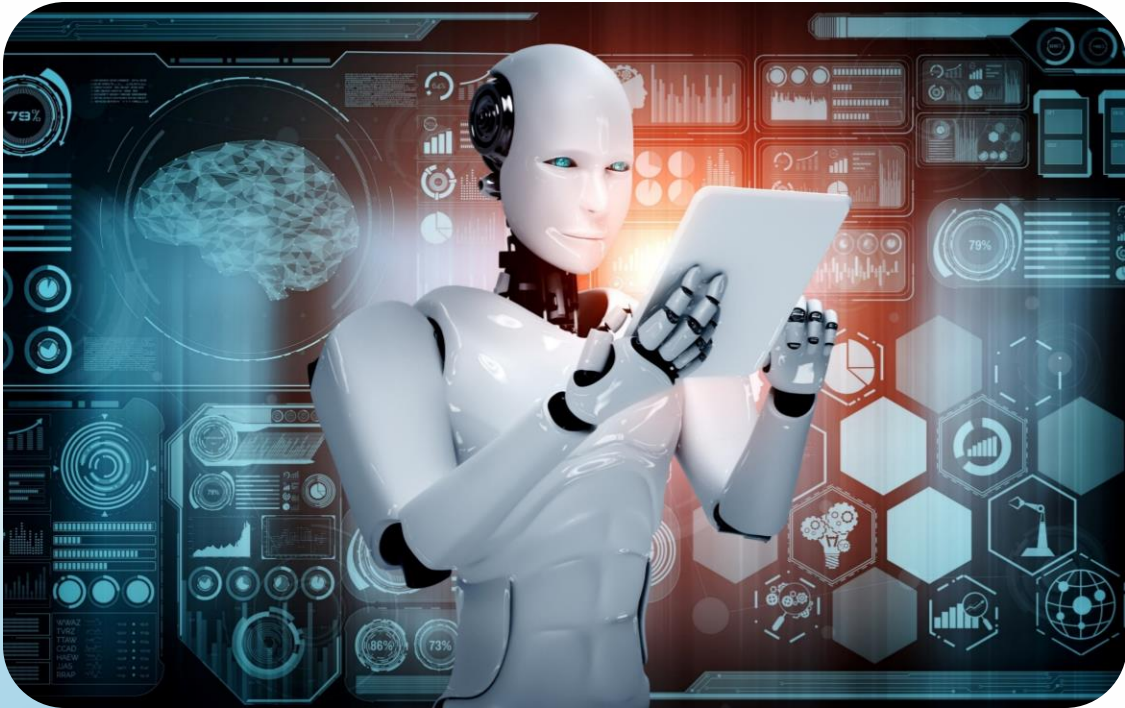


Prioritise reactive agents when rapid response outweighs strategic reasoning



Apply in safety systems and real-time control tasks

Deliberative Agents: Model-Based Planning



- Maintain comprehensive internal models and plan strategically before executing actions
- Simulate potential outcomes and select optimal paths to achieve goals

Deliberative Agents: Model-Based Planning



World model maintenance

Continuously update internal models with obstacles, resources and dynamic conditions



Strategic planning

Simulate "what-if" scenarios, analyse action sequences and optimise for long-term goals



Deliberate execution

Execute with confidence, monitor outcomes and adjust plans from real-world feedback

Deliberative Agents: Industry Applications

Logistics optimisation

Route-planning AI analyses traffic, delivery windows, and vehicle capacity to optimise schedules before dispatch.



Healthcare diagnosis

Medical AI analyses history, symptoms, and test results to recommend treatments aligned with guidelines and drug interactions.



Deliberative Agents: Industry Applications

Key Strengths



Deliver superior
accuracy and
foresight through
comprehensive
analysis



Excel when
strategic decision-
making outweighs
response speed
requirements

Hybrid Agents: Best of Both Worlds



- Combine strategic planning with real-time responsiveness of reactive systems
- Address limitations of pure models to perform reliably in dynamic environments

Hybrid Agents: Best of Both Worlds

Deliberative layer computes optimal path using traffic, weather and destination data.

Updates planning models and reactive responses using experience and feedback

Reactive layer manages obstacles, lane changes and emergency braking

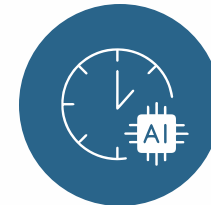
Route planning



Continuous learning



Real-time adaptation



Hybrid Agents: Best of Both Worlds



Autonomous vehicles exemplify hybrid architecture by planning routes strategically while reacting instantly to pedestrians, signals, and road conditions.

Multi-Agent Systems: Distributed Intelligence

System Architecture

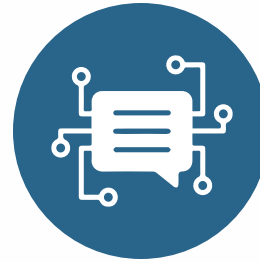
Multi-agent systems deploy specialised agents that coordinate distinct tasks to solve complex problems and achieve shared goals.



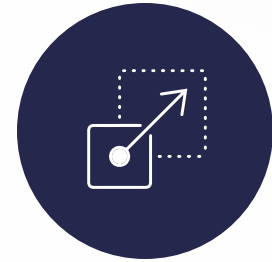
Distributed processing eliminates single points of failure



Specialised agents handle domain-specific tasks



Communication protocols enable coordination



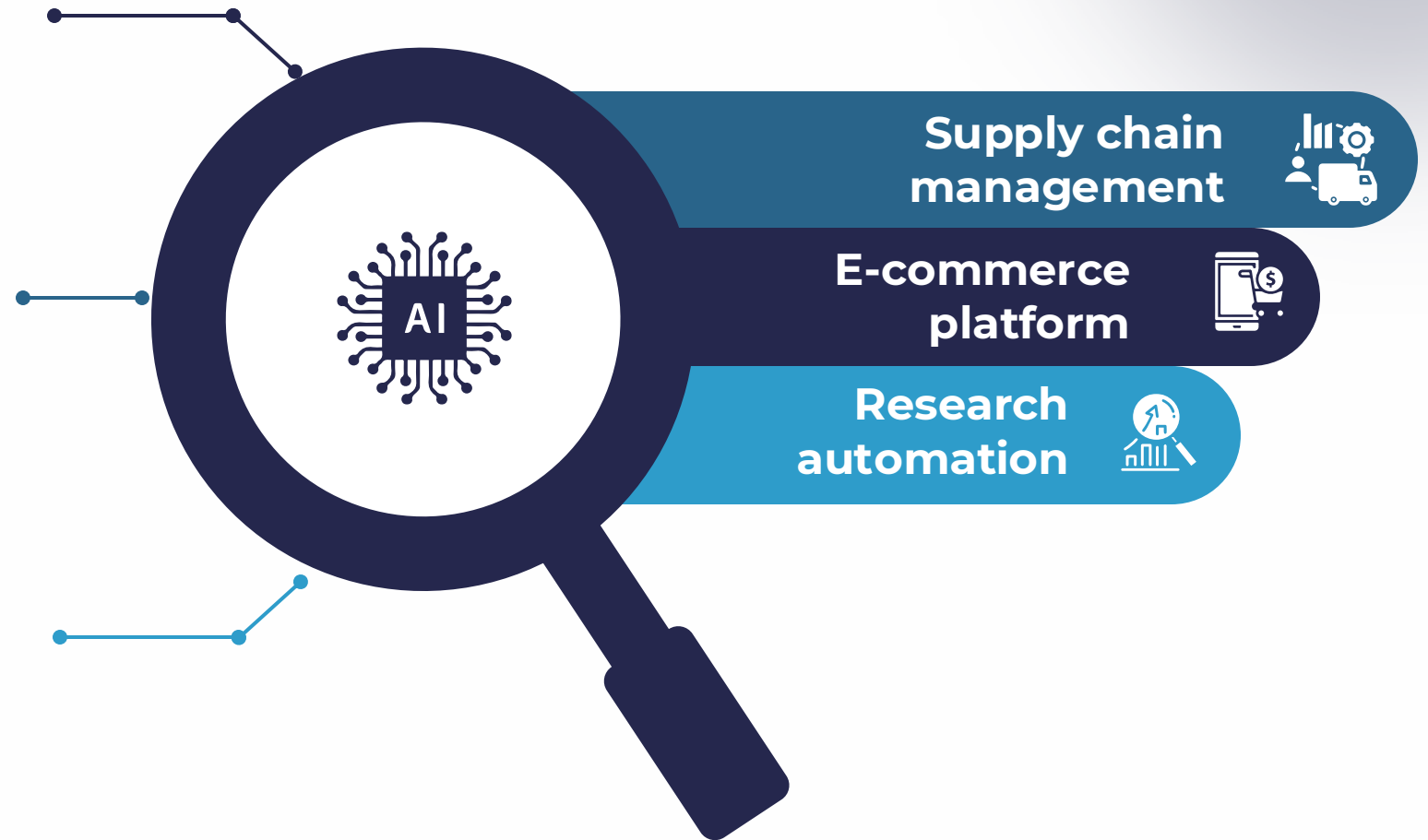
Scalable architecture grows with problem complexity

Multi-Agent Systems: Distributed Intelligence

Agents coordinate inventory management, shipping optimisation and demand forecasting.

Recommendation agent suggests products, inventory agent manages stock, payment agent handles transactions.

Agents read papers, extract insights and draft reports in a coordinated workflow.



Choosing the Right Agent Architecture



<1ms
Reactive response time

Ideal for safety-critical
systems requiring
instant responses



95%
Deliberative accuracy

Superior decision
quality through
comprehensive analysis



3x
Hybrid performance

Industry adoption rate
compared to pure
architectures

Choosing the Right Agent Architecture

Architecture Selection Guide

Reactive

Time-critical
systems (trading
bots, safety
controls)

Deliberative

Strategic
applications
(route planning,
medical
diagnosis)

Hybrid

Complex real-
world systems
(autonomous
vehicles, advanced
robotics)

Multi-agent

Large-scale
distributed
problems
(supply chains,
research teams)

Choosing the Right Agent Architecture

Key Points to Remember



Reactive agents are firefighters — Fast reaction



Deliberative agents are chess players — Strategic planning



Hybrid agents are race car drivers — Plan the race, react to conditions



Multi-agent systems are orchestras — Coordination plus specialisation

Task-Oriented v. Goal-Oriented Agents & Agent Communication Frameworks

Exploring the evolution of intelligent agents from simple task executors to sophisticated goal-driven systems that communicate, reason and adapt autonomously.

The Evolution of Artificial Intelligence

Rule-based systems

Early AI systems followed predetermined rules and decision trees to solve specific problems



Goal-oriented intelligence

Modern agents that pursue broader objectives through dynamic planning and adaptive reasoning



Task-oriented agents

Agents complete specific, well-defined tasks through structured workflows.

Collaborative networks

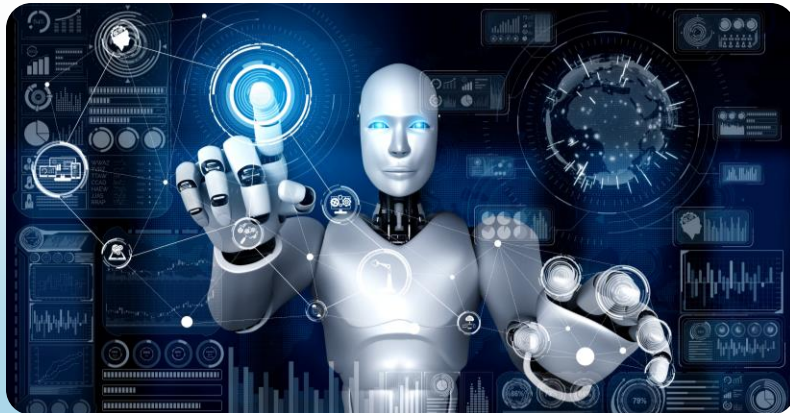
Multi-agent systems communicate and coordinate to achieve complex shared goals.

Defining the Agents

Task-Oriented v. Goal-Oriented

Task-oriented agents

- Complete predefined tasks step-by-step using reactive, rule-based approaches
- Excel at structured workflows but lack flexibility



Goal-oriented agents

- Pursue broader objectives through dynamic planning and adaptive reasoning
- Reason about future states and optimise strategies autonomously



- **Real-world example:** A task-oriented chatbot follows scripts to answer FAQs, while a goal-oriented agent autonomously negotiates and plans workflows.

Task-Oriented Agents in Action



Input recognition

Agent categorises user input using predefined patterns and keywords



Rule application

System applies matching rules or decision trees to select a response



Task execution

Agent executes the structured workflow or delivers a scripted reply



Output delivery

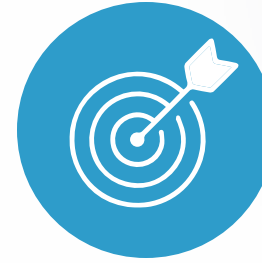
System delivers the output and awaits the next input within task boundaries

Key Features & Differences



Task-Oriented

- **Fixed workflows** — Executes set sequences with minimal deviation
- **Limited reasoning** — Operates within immediate task scope only
- **Reactive behaviour** — Responds via predefined triggers and patterns



Goal-Oriented

- **Dynamic planning** — Adapts strategies by reasoning about future states
- **Memory & learning** — Improves over time using memory and learning systems
- **Proactive pursuit** — Balances objectives, resources, and constraints autonomously

Goal-Oriented Intelligence Framework



Agent Communication

From Message Passing to Goal-Driven Dialogue



Traditional communication

Focuses on message fidelity, bandwidth and reliable data transmission between systems



Goal-oriented communication

Ties messages directly to task success and shared objectives, optimising relevance under resource constraints

"Goal-oriented communication optimises information exchange based on task relevance rather than just message completeness."

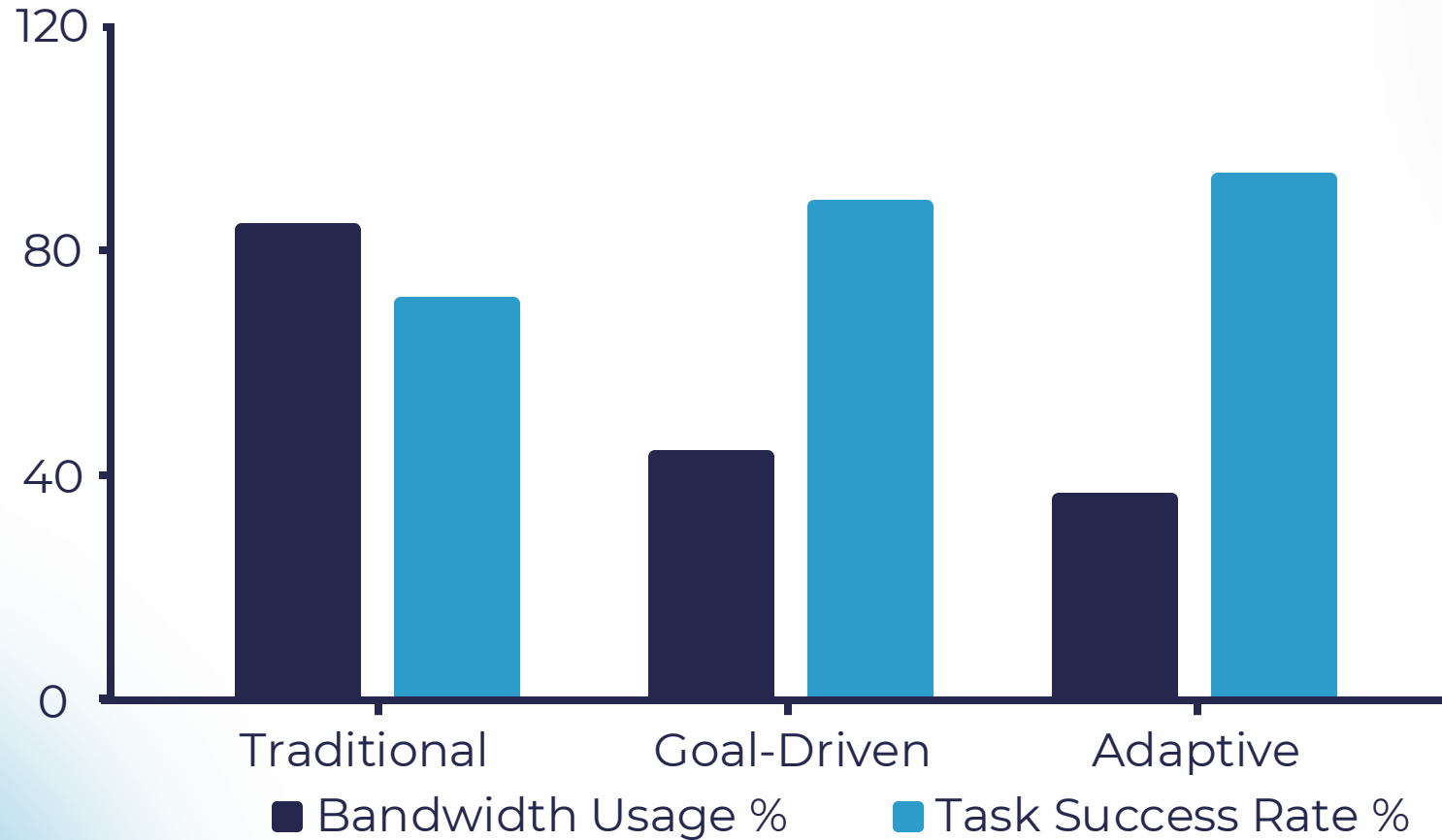
Agent Communication

Example



Swarm robots exchange only goal-critical data, minimising communication while preserving mission effectiveness.

Communication Efficiency Comparison



Goal-driven communication boosts task success while minimising bandwidth, showcasing context-aware efficiency.

Reasoning Frameworks Empowering Agents



Planning algorithms: Use probabilistic models and reinforcement learning to select optimal actions under uncertainty



Memory integration: Integrate tools and memory systems to support complex decision-making and retain knowledge

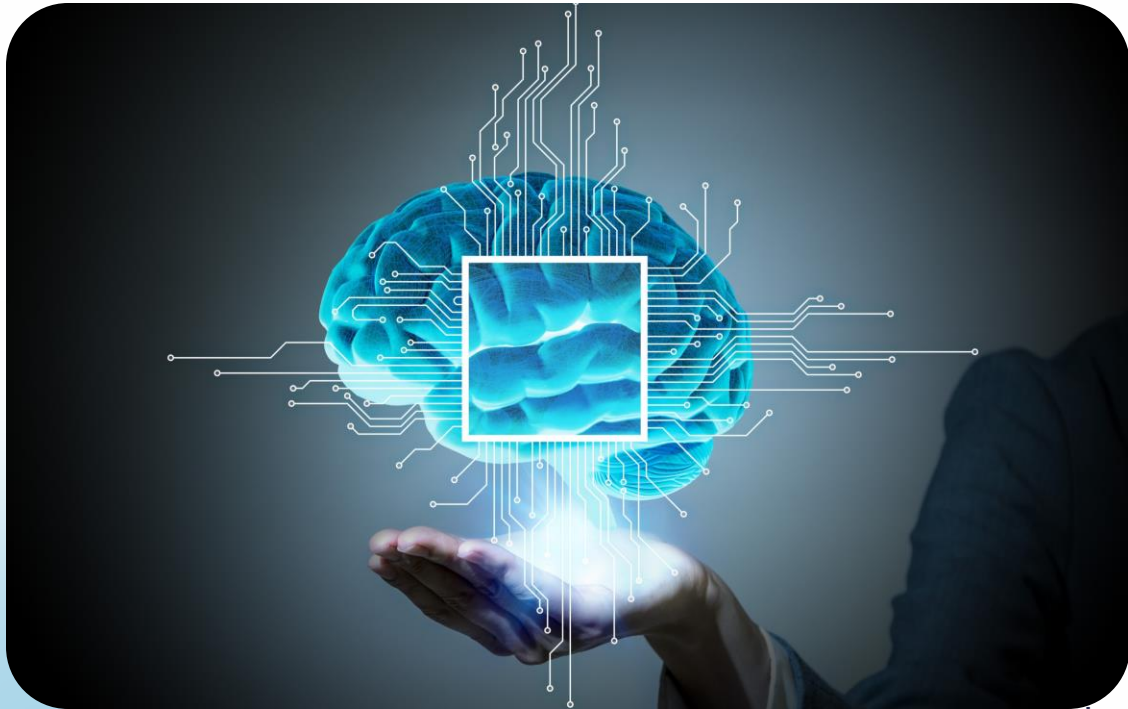


Framework evolution: Employ LangChain and LlamaIndex to build autonomous, goal-driven workflows with enhanced capabilities



Conversational AI: Evolve from reactive bots to proactive, goal-driven dialog systems using advanced models like SocialAGI's GDAD

Breakthrough Technologies Enabling Smart Agents

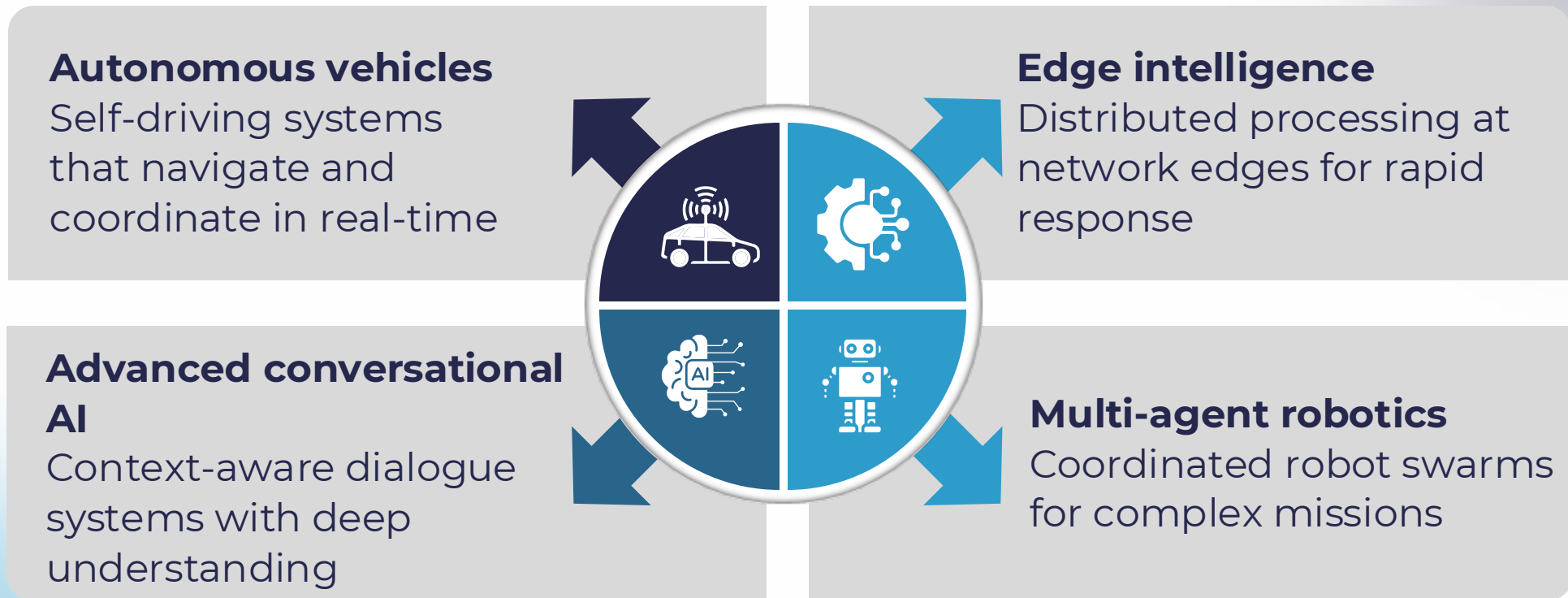


Next-generation agents unify breakthrough technologies to achieve unmatched autonomy and intelligence.

- **Advanced neural networks:** Enable sophisticated reasoning and planning via transformers and attention mechanisms
- **Persistent memory systems:** Maintain context and support learning through long-term and working memory
- **Tool integration frameworks:** Extend agent capabilities beyond language via APIs and external tools

The Future

Dynamic, Collaborative and Context-Aware Agents



The shift from task execution to goal achievement enables smarter, adaptive agents for dynamic, resource-limited environments.

Q & A

Thank you